

OBJECTS (JavaScript)

Easy

1. Create an object student with properties: name, age, course.
2. Access the course property using dot notation.
3. Add a new property grade to an existing object.
4. Update the age of a student object.
5. Delete the course property from an object.

Moderate

6. Loop through all keys in a user object and print key-value pairs.
7. Check if a property email exists in an object.
8. Write a function that accepts an object and returns only the values.
9. Convert an object into an array of keys and values.
10. Prevent modification of an object using JavaScript methods.

Challenging

11. Create a nested object representing an e-commerce order.
12. Safely access a deeply nested property (avoid errors).
13. Write a function to count how many properties an object has.
14. Merge two objects without modifying the originals.
15. Clone an object and prove the original is unchanged.

OBJECT DESTRUCTURING

Easy

16. Extract name and price from a product object.
17. Rename a destructured property while extracting it.
18. Provide default values while destructuring.
19. Destructure an object inside a function parameter.
20. Extract properties from a nested object.

Moderate

21. Destructure only required fields from an API response object.
22. Skip unwanted properties while destructuring.
23. Swap two variable values using destructuring.

- 24. Destructure object returned from a function.
- 25. Extract id and group remaining properties using rest.

Challenging

- 26. Destructure deeply nested objects in one statement.
- 27. Write a function that destructures config options with defaults.
- 28. Destructure props in a React-like function component.
- 29. Extract values from an array of objects using destructuring.
- 30. Use destructuring in a loop (for...of).

SPREAD OPERATOR (...)

Easy

- 31. Create a copy of an array using spread.
- 32. Add a new element to an array without modifying original.
- 33. Copy an object and add a new property.
- 34. Merge two arrays into one.
- 35. Merge two objects into one.

Moderate

- 36. Update a specific property of an object using spread.
- 37. Insert an element at the beginning of an array using spread.
- 38. Clone nested objects and explain shallow copy.
- 39. Use spread to pass array values as function arguments.
- 40. Merge default settings with user settings.

Challenging

- 41. Update an item inside an array of objects immutably.
- 42. Remove an element from an array without splice().
- 43. Create a function that updates user state (React-style).
- 44. Combine multiple objects where later values override earlier ones.
- 45. Explain why spread does not deep copy.

REST OPERATOR (...)

Easy

- 46. Write a function that accepts unlimited numbers and logs them.
- 47. Use rest operator in function parameters.

- 48. Collect remaining values from an array using rest.
- 49. Collect remaining properties from an object.
- 50. Separate first element and rest of an array.

Moderate

- 51. Build a function that calculates average of any number of inputs.
- 52. Create a function that filters even numbers using rest.
- 53. Destructure an object to extract one property and rest.
- 54. Use rest in a function that formats messages.
- 55. Combine rest with destructuring in parameters.

Challenging

- 56. Build a logger function that accepts unlimited messages.
- 57. Write a function that groups remaining user details dynamically.
- 58. Combine rest with reduce to calculate totals.
- 59. Use rest to build a middleware-like function.
- 60. Explain why rest must be the last parameter.

CLASSES (ES6)

Easy

- 61. Create a class Person with name and age.
- 62. Add a method to display person details.
- 63. Create an object from a class.
- 64. Use constructor in a class.
- 65. Add default values in constructor.

Moderate

- 66. Create a Student class that extends Person.
- 67. Override a method in a child class.
- 68. Use super() correctly.
- 69. Add getters and setters.
- 70. Create a static method.

Challenging

- 71. Build a BankAccount class with deposit & withdraw.
- 72. Add validation inside class methods.

- 73. Implement encapsulation using private fields.
- 74. Explain class vs constructor function.
- 75. Design a mini class-based system (library, school, store).

Bonus: Mixed Scenario Questions (Interview Style)

- 76. Update React state using object spread – why is it important?
- 77. Use destructuring in API response handling.
- 78. Combine rest and spread in one function.
- 79. Convert ES5 object code into ES6.
- 80. Identify and fix mutation bugs in code.